1、设 $M(\lambda)N(\lambda)=N(\lambda)M(\lambda)=I$，$N(\lambda)=N_n\lambda^n+N_{n-1}\lambda^{n-1}+\cdots+N_0$

$\therefore M_0N_0=I$ $\therefore M_0$ 非奇

2、若 $Q(\lambda)$，R 不唯一，则存在 $Q'(\lambda)$，R'，使

$$M(\lambda)=(\lambda I-B)Q(\lambda)+R=(\lambda I-B)Q'(\lambda)+R'$$

$$(\lambda I-B)(Q(\lambda)-Q'(\lambda))=R'-R$$

对比次数，$Q(\lambda)=Q'(\lambda)$ $R=R'$

3、不妨设 P 可逆

充分：$P^{-1}(\lambda I_n-A)P=\lambda I_n-B$ $P^{-1}AP=B$ $\therefore \lambda I_n-A$ 与 λI_n-B 相抵

必要：$\because \lambda I_n-A$ 与 λI_n-B 相抵 $\therefore A,B$ 相似 $\therefore B=P^{-1}AP$

设 $Q=P^{-1}A$ $\therefore B=QP$ $A=PQ$

4、对角矩阵 $\mathrm{diag}\{f_1(\lambda),f_2(\lambda),\cdots f_n(\lambda)\}$ 先对换 i,j 行再对换 i,j 列，可实现 $f_i(\lambda)$，$f_j(\lambda)$ 互换

而对任一排列可通过作若干次对换实现，故 λI_n-A 与 λI_n-B 相抵 $\therefore A$ 与 B 相似

1、(1) $\lambda I-A=\begin{pmatrix}\lambda-1 & -2 & 1\\ -1 & \lambda-1 & -1\\ \lambda+2 & \lambda+2 & \lambda-1\end{pmatrix}\to\begin{pmatrix}1 & -2 & \lambda-1\\ -1 & \lambda-1 & -1\\ \lambda+2 & \lambda+2 & \lambda-1\end{pmatrix}\to\begin{pmatrix}1 & -2 & \lambda-1\\ 0 & \lambda-3 & \lambda-1\\ 0 & 2\lambda+4 & -(\lambda-1)(\lambda+2)\end{pmatrix}\to\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda-3 & \lambda-1\\ 0 & 2\lambda+4 & -(\lambda-1)(\lambda+2)\end{pmatrix}\to$

$\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & \frac{1-\lambda}{2}\\ 0 & (\lambda+2)(\lambda-1) & -\frac{(\lambda+2)(\lambda-1)}{2}\end{pmatrix}\to\begin{pmatrix}1 & & \\ & 1 & \\ & & (\lambda-1)^2(\lambda+2)\end{pmatrix}$

(2) $\lambda I-A=\begin{pmatrix}\lambda+3 & -1 & -1\\ 3 & \lambda & -2\\ 2 & -1 & \lambda\end{pmatrix}\to\begin{pmatrix}1 & \lambda+3 & -1\\ -\lambda & 3 & -2\\ 1 & 2 & \lambda\end{pmatrix}\to\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda^2+3\lambda+3 & -\lambda-2\\ 0 & -\lambda-1 & \lambda+1\end{pmatrix}$

$\to\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda+1 & 0\\ 0 & 0 & (\lambda+1)^3\end{pmatrix}$

(3) $\lambda I-A=\begin{pmatrix}\lambda & -1 & -1\\ 0 & \lambda-1 & 0\\ \lambda+1 & -1 & \lambda-2\end{pmatrix}\to\begin{pmatrix}1 & \lambda & -1\\ 1-\lambda & 0 & 0\\ \lambda+1 & 1 & \lambda-2\end{pmatrix}\to\begin{pmatrix}1 & 0 & 0\\ 1-\lambda & \lambda^2-\lambda & 1-\lambda\\ -1 & -\lambda^2+1 & \lambda-1\end{pmatrix}$

$\to\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda-1 & 0\\ 0 & 0 & (\lambda-1)^2\end{pmatrix}$

2、(1) $A(\lambda)\to\begin{pmatrix}1 & 2\lambda-1 & \lambda^2\\ 0 & \lambda^2 & -\lambda\\ 1 & \lambda^2+\lambda-1 & -\lambda^2\end{pmatrix}\to\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda^2 & 0\\ 0 & \lambda^2-\lambda & 0\end{pmatrix}\to\begin{pmatrix}1 & 0 & 0\\ 0 & \lambda^2 & 0\\ 0 & -\lambda & \lambda^2\end{pmatrix}$

$\to\begin{pmatrix}1 & & \\ & \lambda & \\ & & \lambda^3+\lambda\end{pmatrix}$

(2) $A(\lambda)\to\begin{pmatrix}-\lambda & & \\ & \lambda^2-\lambda & \\ & & \lambda^2\end{pmatrix}\to\begin{pmatrix}-\lambda & & \\ & (\lambda-1)^2 & \\ & & \lambda^2(\lambda-1)\end{pmatrix}\to\begin{pmatrix}\lambda & & \\ & \lambda-1 & \\ & & \lambda(\lambda-1)^2\end{pmatrix}$

$\to\begin{pmatrix}1 & & \\ & \lambda(\lambda-1) & \\ & & \lambda^2(\lambda-1)\end{pmatrix}\to\begin{pmatrix}1 & & \\ & \lambda(\lambda-1) & \\ & & \lambda^2(\lambda-1)\end{pmatrix}$

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3. $\begin{pmatrix} g(\lambda) \\ f(\lambda) \end{pmatrix} \Rightarrow$

设 $\because f(\lambda)v(\lambda)+g(\lambda)u(\lambda)=d(\lambda)$ $\therefore$

$\therefore \begin{pmatrix} f(\lambda) & g(\lambda) \end{pmatrix} \to \begin{pmatrix} f(\lambda) \\ f(\lambda)v(\lambda) & g(\lambda) \end{pmatrix} \to \begin{pmatrix} f(\lambda) \\ d(\lambda) & g(\lambda) \end{pmatrix} \to \begin{pmatrix} d(\lambda) & \frac{f(\lambda)g(\lambda)}{d(\lambda)} \end{pmatrix} \Rightarrow \begin{pmatrix} d(\lambda) \\ m(\lambda) \end{pmatrix}$

$\therefore \begin{pmatrix} f(\lambda) \\ g(\lambda) \end{pmatrix}$ 与 $\begin{pmatrix} d(\lambda) \\ m(\lambda) \end{pmatrix}$ 相抵，$\begin{pmatrix} g(\lambda) \\ f(\lambda) \end{pmatrix}$ 同理

充分：取 $A^*(\lambda)$ $A(\lambda)A^*(\lambda)=|A(\lambda)|I_n=A^*(\lambda)A(\lambda)$

$\therefore |A(\lambda)|^{-1}A^*(\lambda)$ 即为 $A(\lambda)$ 的逆 $\therefore A(\lambda)$ 可逆

必要：设 $A(\lambda)B(\lambda)=I_n$ $\therefore |A(\lambda)||B(\lambda)|=|I_n|=1$

$\therefore \deg|A(\lambda)|+\deg|B(\lambda)|=0$ $\therefore |A(\lambda)|$ 为非零常数

设 $P(\lambda)(\lambda I-A)Q(\lambda)=\operatorname{diag}\{1,\dots,d_1(\lambda),\dots d_m(\lambda)\}$

$\therefore |P(\lambda)||\lambda I-A||Q(\lambda)|=d_1(\lambda)\cdots d_m(\lambda)$

设 $|P(\lambda)||Q(\lambda)|=c$ $\therefore c|\lambda I-A|=d_1(\lambda)\cdots d_m(\lambda)$

又 $\because |\lambda I-A|, d_1(\lambda)\cdots d_m(\lambda)$ 首一 $\therefore c=1$

$\therefore |\lambda I-A|=d_1(\lambda)\cdots d_m(\lambda)$

Date

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